

## **JOBSHEET 1**

### **Programming Language (Java Development Kit/JDK Installation)**

*All of Experiments 1 to 3 and the Assignment in this jobsheet are to be completed in the classroom/laboratory together with the lecturer and the lab assistants during the face-to-face session (total allocation of 6 credit hours/practicum hours), not as independent work outside of class.*

#### **1. Objectives**

- Students are introduced to the concepts of programs and programming languages
- Students are able to install Java programming tools
- Students understand the basic structure of Java
- Students are able to compile and debug programs

#### **2. Experiments**

Before starting the practicum, it is important to understand the following basic concepts. A programming language is a set of rules (syntax) used to write instructions that can be executed by a computer; one of the programming languages that will be used in this course is Java. Java is platform-independent, meaning that the same program code can be run on various operating systems (Windows, macOS, Linux) as long as a Java Virtual Machine (JVM) is available for it. In order for the Java program code we write (with the .java extension) to be executed by the computer, the code must first be translated by the compiler (the javac command) into bytecode (with the .class extension); only then is that bytecode executed using the java command. The Java Development Kit (JDK) is a set of tools containing the compiler and other components needed to develop and run Java programs, so the first step before writing code is to install the JDK.

##### **2.1 Experiment 1: Installing the Java Development Kit/JDK**

**Experiment time : 120 minutes**

1. Download the JDK by opening the following address:

<https://www.oracle.com/java/technologies/downloads/>

2. Double-click the installation file you have downloaded, then follow the installation instructions.
3. The next step is setting the PATH (for the Windows operating system only), so that the java command can be recognised.
4. To set the PATH, open **Control Panel-> System-> Advanced System Setting-> Environment Variable**. Then look for the PATH variable; if it does not exist yet, create the PATH variable.
5. The next step is filling in the PATH variable. If the PATH variable already has a value, do not delete the existing value but add **C:\Program Files\Java\jdk\bin** ( use the ; character as the separator between values).  
*NB: Adjust the JDK location you add to the folder where Java is installed; in the example above the JDK is installed in C:\Program Files.*
6. Open Command Prompt (Win+R, then type cmd), then type the command **javac**. If the command is recognised, the Windows operating environment already supports Java programs; but if it is not yet recognised, check the PATH setting (there may be an error when entering the location of the bin directory in the PATH variable).

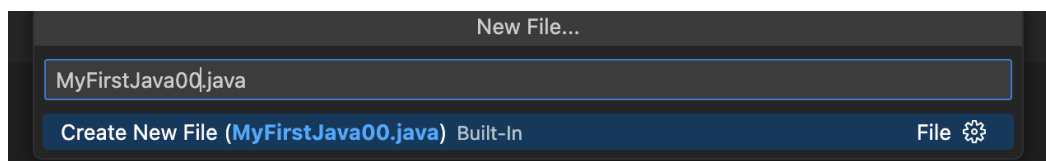
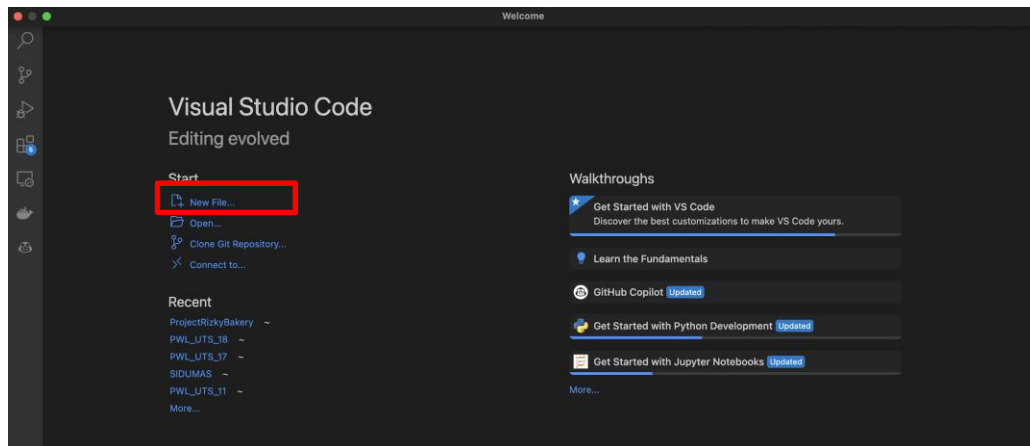
## Questions

1. Explain the purpose of adding the location of Java's bin folder to the PATH variable!
2. Explain the purpose of the javac command when entered in the command prompt!

## 2.2 Experiment 2 : Basic Java Structure

### Experiment time : 70 minutes

1. Download Visual Studio Code from the following page:  
<https://code.visualstudio.com/download>
2. Double-click the installation file you have downloaded, then follow the installation instructions
3. Open Visual Studio Code that you installed in the previous step, then create a new file and save it with the name MyFirstJavaStudentNumber.java (use your own student/roll number).



*\*replace 00 with your own student/roll number*

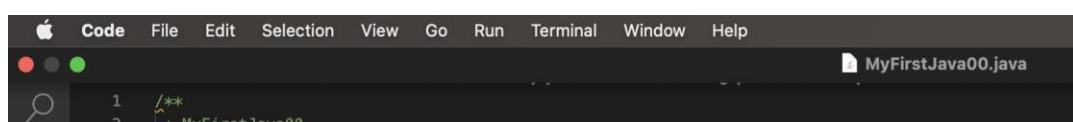
4. Write the following program code: (the following program code is the basic structure of Java). Replace 00 with your student/roll number.

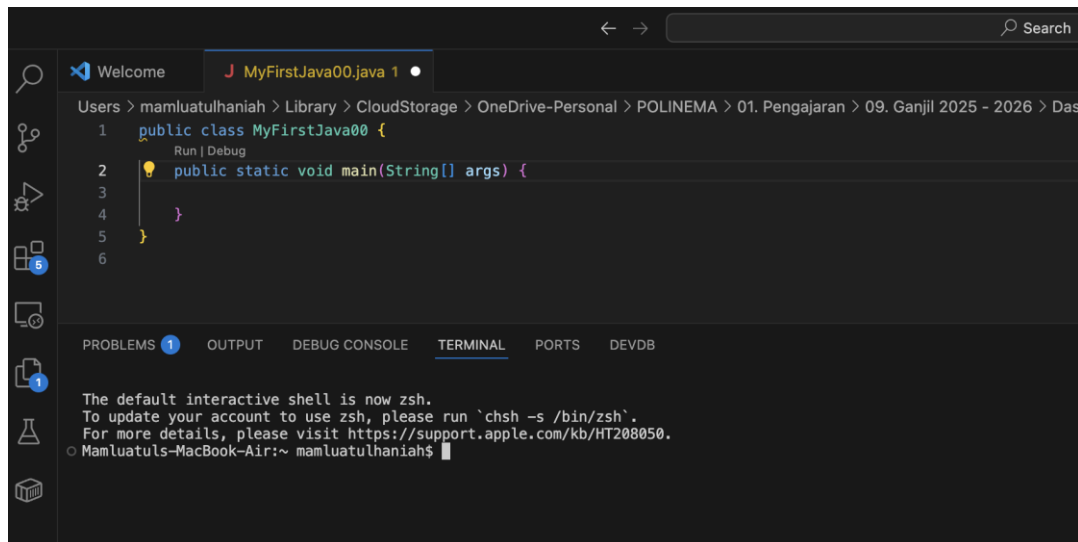
```
public class MyFirstJava00 {
    public static void main(String[] args) {

    }
}
```

*Description:*

- public is a keyword in Java indicating that an object, method, or attribute can be accessed from another class.
  - class is a keyword in Java used to create a class
  - static is a keyword used so that a method does not need to be instantiated first
  - void is a keyword used so that a method does not return any value, i.e. returns nothing
  - The class name and the file name must be the same
5. Save the program code (File-> Save)
  6. Open the terminal in Visual Studio Code





7. In the terminal you must navigate to the directory where you saved the **MyFirstJava00.java** file. To change directory, you can use the `cd` command. For example, if the **MyFirstJava00.java** file is saved in “C:/Users/mamluatulhaniah/Library/CloudStorage/OneDrive-Personal/POLINEMA/01. Pengajaran/11. Ganjil 2026-2027/Dasar Pemrograman/01. Pertemuan 1”, the example is as follows:

```
Mamluatuls-MacBook-Air:~ mamluatulhaniah$ cd
"/Users/mamluatulhaniah/Library/CloudStorage/OneDrive-Personal/POLINEMA/01.
Pengajaran/09. Ganjil 2025 - 2026/Dasar Pemrograman/01. Pertemuan 1"
```

Directory where the java file is stored

**Note:** Adjust this to your own directory

8. Once you are inside the folder where the **MyFirstJava00.java** file is saved, the next step is to compile it by typing the command `javac FileName.java` (`javac MyFirstJava00.java`)

```
Mamluatuls-MacBook-Air:01. Pertemuan 1 mamluatulhaniah$ javac MyFirstJava00.java
Mamluatuls-MacBook-Air:01. Pertemuan 1 mamluatulhaniah$
```

9. If the compilation succeeds and there are no errors, a new file named **MyFirstJava.class** will appear (open and check the folder where you saved the **MyFirstJava.java** file).

|                     |             |           |                  |
|---------------------|-------------|-----------|------------------|
| MyFirstJava00.class | Today 00:57 | 271 bytes | Java class file  |
| MyFirstJava00.java  | Today 00:52 | 86 bytes  | Java so...e code |

10. To run the compiled program, enter the command `java ClassName` (`java MyFirstJava00`)

```
Mamluatuls-MacBook-Air:01. Pertemuan 1 mamluatulhaniah$ java MyFirstJava00
Mamluatuls-MacBook-Air:01. Pertemuan 1 mamluatulhaniah$
```

11. In the experiment you have carried out, no output is displayed on the terminal screen yet, because the program code written is only the basic structure of a Java program.
12. Modify the MyFirstJava00.java file by adding the following code!

```
public class MyFirstJava00 {  
    public static void main(String[] args) {  
  
        System.out.println("Hello, my name is XXXXX!");  
    }  
}
```

**Note** : replace xxxx with your name

13. Save the file (File->Save)
14. Compile it as in step 8.
15. If there are no errors, perform step 9.
16. Observe the output of the program! Is it the same as follows:

```
Mamluatuls-MacBook-Air:01. Pertemuan 1 mamluatulhaniah$ java MyFirstJava00  
Hello, my name is XXXXX!
```

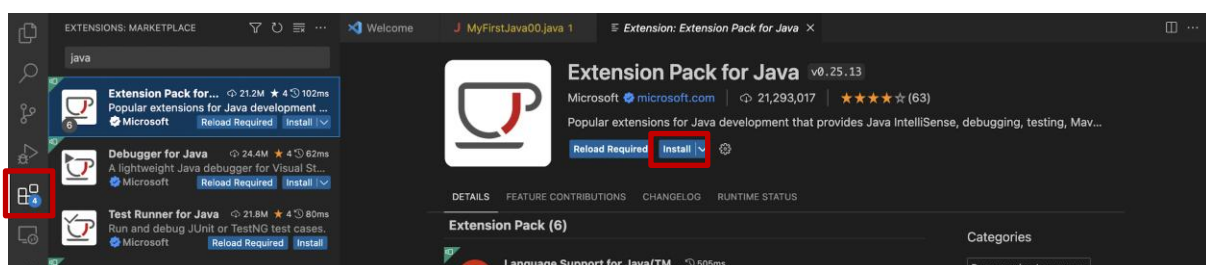
## Questions!

1. Explain the function of the command javac MyFirstJava00.java in the experiment above!
2. Explain the function of the command java MyFirstJava00 in the experiment above!

## 2.3 Experiment 3 : Adding an Extension in Visual Studio Code

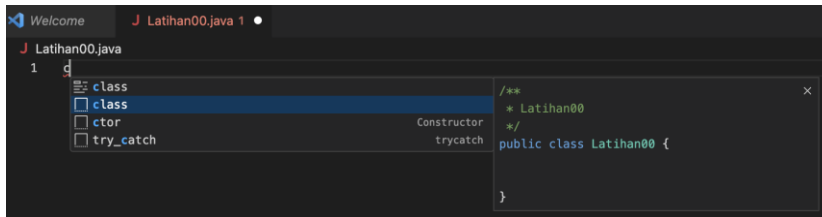
**Experiment time : 60 minutes**

1. Open the Extension menu, then search for “java”; the Extension Pack for Java will appear, and click install



2. Create a new file named Latihan00.java, replacing 00 with your own student/roll number
3. Write the basic Java structure. The extension pack will offer recommendations when you are about to create a class. In the following example, typing c will bring up a recommendation for creating a class. When the recommendation appears, you simply

press the tab key. The same applies to the main function: type main and press tab, and the main function will be created automatically



```
1  /**
2   * Latihan00
3   */
4  public class Latihan00 {
5
6      public static void main(String[] args) {
7
8      }
9  }
```

4. Add the following code:

```
1  public class Latihan00 {
2
3      public static void main(String[] args) {
4          System.out.println("Hello, my name is XXXXX!");
5
6          System.out.println("Learning Java is easy!");
7          System.out.println("The more often you practise, the better you
get.");
8      }
9  }
```

5. Run the code above by pressing the triangular run button in the top right corner



6. Observe the output of the program! Is it the same as the following image:

```
Hello, my name is XXXXX!
Learning Java is easy!
The more often you practise, the better you get.
```

## Questions!

1. Change the System.out.println command to System.out.print on lines 6 and 7, then run the program!
2. What is the difference between System.out.println("....."); and System.out.print(".....");? Explain!
3. Change the commands on lines 6 and 7 to the following :

```
System.out.print("\nLearning Java is easy!");
System.out.print("\nThe more often you practise, the better you get.");
```
4. Explain the function of the \n command on lines 6 and 7!

5. On line 7, add the characters `//` so that the line becomes

```
//System.out.print("\n\nThe more often you practise, the better you get.");
```

6. Run the program and observe what happens! Explain the function of the `//` characters

## 1. Assignment

**Working time : 30 minutes (completed in the classroom/laboratory together with the lecturer and assistants, not as independent work at home)**

Consider the following problem (adapted from the Bebras Challenge – Robot Arm, from Germany). A class at Beaver High School built a prototype robot arm and then tested it on a table with 3 trays (A, B, C). Initially: tray A contains 1 star ball, tray B contains 1 moon ball, and tray C is empty. The robot arm then executes the following steps in order:

- 1) Take the ball in tray A and put it in tray C.
- 2) Take the ball in tray B and put it in tray A.
- 3) Take the ball in tray C and put it in tray B.

Question: After the 3 steps above have been executed, which of the statements below are correct (more than one may apply)? (a) The two balls have swapped places; (b) there are two balls in tray A; (c) there are two balls in tray B; (d) tray A is empty; (e) tray C is empty; (f) nothing has changed, each ball has returned to its original place.

Write a Java program (for example named RobotArm00.java, replacing 00 with your student/roll number) that prints the contents of trays A, B, and C after EVERY step (Step 1, Step 2, Step 3), ending with a conclusion line that prints which statement is correct according to your own analysis. Example output format (the dots are to be filled in based on your own analysis, not a ready-made answer):

```
MamluatulS-MacBook-Air:~ mamluatulhaniah$ java RobotArm00
After Step 1: Tray A = ..., Tray B = ..., Tray C = ...
After Step 2: Tray A = ..., Tray B = ..., Tray C = ...
After Step 3: Tray A = ..., Tray B = ..., Tray C = ...
Conclusion: The correct statement is "...
MamluatulS-MacBook-Air:~ mamluatulhaniah$
```